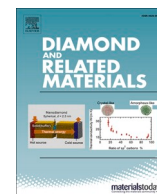




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Novel synthesis of carbon nanofiber aerogels from coconut matrix for the electrochemical detection of glucose

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ABSTRACT

Bacterial cellulose aerogels (BCAs) were obtained from a delicious food grade coconut. Following with a simple process of carbonized calcination (700 °C~1100 °C). The obtained coconut matrix carbon nanofiber aerogels (CMCNFs) exhibit low density ($\leq 3.47 \text{ mg/cm}^3$), high surface area ($\geq 1102 \text{ m}^2/\text{g}$), and high porosity ($\approx 99.6\%$), which are much higher than those of common BCAs owing to the removal of hydrogen, oxygen and other elements along with only small volume changes. Due to the flexibility of the nanofiber matrix of our presoma, the three-dimensional network micro-structure is preserved during high temperature carbonization process. The obtained CMCNFs show an excellent conductivity bearing a direct electron transfer process between glucose oxidase (GOx) and electrode surface in the electrochemical processes. It was discovered that CMCNFs can achieve a fast direct electron transfer (DET) of GOx to the rapid detection of glucose. This character should be related to the three-dimensional pore structure determined by carbonization temperature, which has been proved that the CMCNFs calcined at 1000 °C possesses the best capabilities. With merits above, a highly sensitive glucose sensor is successfully prepared by the simple carbonization of coconut cellulose matrix, which expands the application scope of bacterial cellulose aerogels. This work also offers insights for the future develop of protein-based electrochemical devices.

1. Introduction

Porous carbon materials have been widely used in catalysts [1–3], adsorbents [4,5] and gas storage [6–8], in regard of their high specific surface area, porosity, excellent electrical/thermal conductivity, and adjustable surface properties. Common synthesis approaches for porous carbon materials usually include hard template method, soft template method and activation method [9]. However, process difficulties including high cost, complex synthesizing procedures and low productivity are hurdling the extensive application of porous carbon materials, which further obstructs the material commercialization. Recently, carbon nanofiber aerogels (CNFs) have attracted wide attention as a new porous carbon materials [10–12]. It is a new lightweight porous multi-functional aerogel with a porosity of 80% to 99% and a specific surface

area of 1500–2500 m^2/g , also its surface/pore structure can be adjusted through synthesizing processes [13,14]. Comparing with traditional inorganic aerogel such as silica aerogels [15,16], CNFs which possess a higher electrochemical conductivity are more capable of the application of supercapacitors [17–19]. Furthermore, practical applications of CNFs are still extremely limited due to the complex and expensive production methods [20,21]. Therefore, synthesis pathways of CNFs with characters of environmental protection, simple procedure and easy source materials are urgently required.

In fact, the novel cellulose aerogels based on nanofibrillated cellulose have aroused a wide research interests owing to their good biocompatibility, high specific strength and modulus, and environmental-friendly features [22–26]. Also that CNFs can be obtained simply through high temperature calcination of the cellulose matrix of bacterial

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cellulose aerogels (BCAs) [27]. It became a distinctive material in the fabrication of carbon aerogels. Currently, three raw materials have been reported to fabricating cellulose based carbon aerogels, namely (i) plant cellulose, featured excellent mechanical properties, especially soft texture and high strength cotton cellulose, which possesses the best product purity and performance, a promising thermal stability [28–30]; (ii) marine biological cellulose, for example seaweed cellulose with great surface ratio, good toughness and controllable biosynthesis [31–33]; (iii) bacterial cellulose, a uniformed structure with diameter of about 10–100 nm and good biocompatibilities, shape maintenance, water permeability, high crystallinity and good tensile strength [34,35]. Most of the plants and marine organisms need to be separated and purified through complex extraction and purification process before utilizing, as to the third kind of bacterial cellulose and other related biomass materials, they can be directly adopted in the preparation of cellulose based carbon aerogels. Moreover, cellulose based carbon aerogels benefit the sustainable development of nanomaterials and carriers. The simple synthesizing procedure makes them valuable in the development and application of environmental protection materials. It has demonstrated that CNFs was one of the best matrix to well trap or immobilize enzymes for biosensing platform [36], and it has been investigated and used in electrochemical biosensors. In more details, the network of CNFs can be controlled by their diameter for optimal biosensor performance; Palladium doped CNFs biosensor can simultaneously and sensitively detect hydrogen peroxide and glucose [37]. In addition, CNFs have also exhibited the direct electron transfer (DET) of heme-proteins [38]. In brief, porous CNFs was an excellent material for superior biosensors owing to the high specific surface area, fast reactivity, good conductivity, and strong immobilization ability.

Duo to the fast response and high sensitivity of glucose sensor made from CNFs. In this work, coconut matrix bacterial cellulose was adopted to produce porous carbon nanofibers aerogels (CMCNFs). The product was simply prepared through high temperature heat treatment (700 °C–1100 °C). Here, the highly sensitive glucose detector was synthesized and discovered that the sensitivity of the constructed sensor showed difference in various carbonization temperatures, which indicates a key factor for the selection of high sensitivity CMCNFs for glucose detection. These discovered results on the application of highly sensitive glucose detection make a significant impact for the designing of future protein-based electrochemical devices.

2. Experimental

2.1. Chemicals

Glucose oxidase from *Aspergillus niger* (GOx) was purchased from Sigma Aldrich Co. Ltd., tert-butanol ($C_4H_{10}O$), anhydrous ethanol ($EtOH$, C_2H_5OH), were purchased from Shanghai Aladdin Biochemical Technology Co. Ltd., China. Sodium hydroxide (NaOH) and Hydrogen chloride (HCl) were purchased from Sinopharm Chemical Reagent Co. Ltd., China. 0.05 M phosphate buffer solution (PBS, pH 7.0) was prepared by Na_2HPO_4 purchased from Shanghai Aladdin Biochemical Technology Co. Ltd. The pH was adjusted with 1.0 M HCl and 2.0 M NaOH. All chemicals were of analytical grade and used without further purification. Deionized water (H_2O) is ultrapure water (18.2 M Ω /cm) treated by UHP ultrapure water machine.

2.2. Experiment methods

2.2.1. Synthesis of BCAs

BCAs was obtained from a delicious food grade coconut via freeze-drying after alkali treatment according to a procedure previously published by our group [39–41]. This process can be divided into three steps as follows: Firstly, the grade coconut was immersed in deionized water (4 h) to wash the sugar out. Secondly, the washed sample was heated to 90 °C in a NaOH solution with mass fraction of 4% (6 h) shown in

Fig. S1, then washed with deionized water (10 h); during this period, change the washing solution every half an hour until the pH of the washing solution is 7. Finally, it was replaced by the deionized water with a mixture of deionized water and tert-butanol (ensure the volume of the sample remains unchanged after freezing), then freeze dried (20 h) to obtain BCAs.

2.2.2. Synthesis of CMCNFs

The as-obtained BCAs were then pyrolyzed under a flowing argon atmosphere at 7 °C min⁻¹ to 700 °C for 1 h and then at 5 °C min⁻¹ to each of 800–1100 °C and keep the temperature at the end point for 2 h to yield black CMCNFs.

2.3. Materials characterization

Powder X-ray diffraction (XRD) patterns were obtained on an Rigaku Smartlab diffractometer with a Cu K α radiation source ($\lambda = 1.5418 \text{ \AA}$). Field-emission scanning electron microscopy (SEM) characterization of the aerogels was performed by a Hitachi SU 4800 (Japan) with a 10 kV accelerating voltage. Before SEM imaging and an EDX line scan were performed on Tecnai G2 F20 S-TWIN, operated at 200 kV. A transmission electron microscope (TEM Hitachi HT 7700 Japan) was used to further test the nanostructure of the samples. Nitrogen sorption isotherms were measured at 77 K using a Micromeritics ASAP 2020 analyzer. The sample was degassed at 120 °C for 2 h before testing. Brunauer-Emmett-Teller (BET) analysis from the amount of N_2 absorbed at various relative vapor pressures (six points $0.05 < P/P_0 < 0.3$, nitrogen molecular cross-sectional area = 0.162 nm²) was used to determine the surface area. Chemical properties of porous materials were characterized by Fourier transform infrared spectroscopy (FTIR Bruker TENSOR II Germany).

2.4. Electrochemical measurements

Electrochemical properties of porous materials were characterized by using CHI 760E electrochemical workstation. Electrochemical studies were performed in a three electrode electrochemical cell using modified GCE as a working electrode (diameter 5 mm), Ag/AgCl electrode (Sat. KCl) as a reference electrode and platinum wire as a counter electrode. All measurements were carried out at ambient temperature and all the solutions were deoxygenated by bubbling pre-purified N_2 gas for 20 min and a N_2 atmosphere environment was kept in whole electrochemical measurements (except glucose catalysis). For the electrochemical test, the working electrode was prepared as below: the GCE with 5 mm diameter was polishing with 0.05 μ m alumina slurry using a Buehler polishing kit, then ultrasonicated in water for 2 min and dried before use. 10 μ L of CMCNFs was deposited on the pre-cleaned GCE and drying at room temperature. Then 10 μ L of GOx (10 mg mL⁻¹) was deposited on the surface of CMCNFs modified GCE and dried at ambient temperature. Then porous CMCNFs/GOx modified GCE was washed with buffer to remove loosely adsorbed enzyme. Finally 5 μ L nafion solution (5%) was deposited and dried at room temperature. For comparison, GOx and CMCNFs modified GCEs were prepared following the same procedure.

3. Results and discussion

3.1. Schematic diagram of glucose sensor based on bio cellulose

CMCNFs were conveniently obtained by high temperature carbonization of cellulose nanofibers of BCAs and applying to direct electrochemistry of GOx for rapid detection of glucose as shown in Fig. 1. In this work, BCAs were prepared through a purification of the food grade coconut (Fig. S1). It should be noted that CMCNFs can be easily loaded with GOx, and can be directly used for glucose detection due to its good conductivity. Specifically, GOx consists of two identical polypeptide chains and one flavin adenine dinucleotide (FAD/FADH₂) redox center,

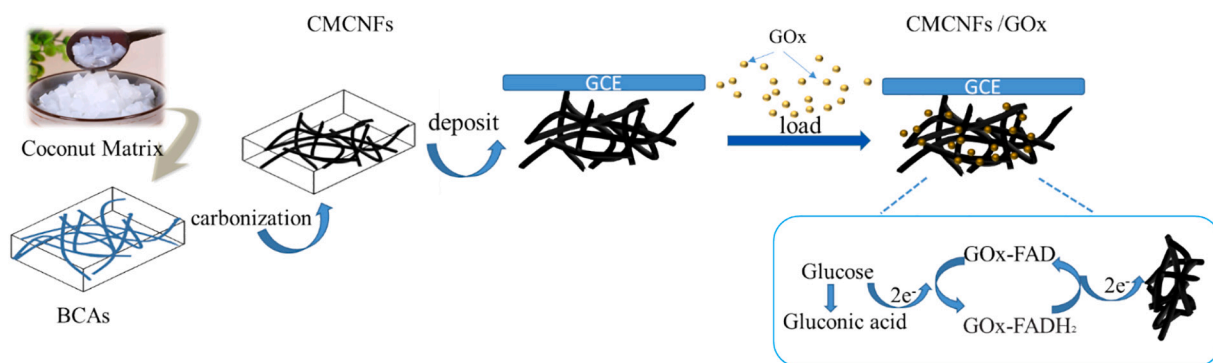


Fig. 1. Schematic illustration of CMCNFs electrode materials realizing the direct electrochemistry of GOx.

which is widely used as commercial glucose sensors because of high sensitivity and selectivity. Comparing with the Clark glucose enzyme sensor [42,43], and the artificial mediator-based glucose sensors [44–46], DET-involved highly sensitive sensors [47] of CMCNFs as biosensing platform greatly improved the efficiency of electron transfer, and the fast-direct electron transfer (DET) involved sensor has the highest sensitivity and selectivity due to its high energy conversion efficiency.

Previous studies have shown that CNFs were better platform carrier for biosensor [48], for which improved DET process between FAD and electrode. Herein, there is no doubt that the simple and rapid preparation of CMCNFs show potential for rapid electrochemical detection of glucose.

3.2. Morphology of BCAs and CMCNFs

Fig. 2a shows bulk BCAs and CMCNFs after high temperature carbonization and the textural properties of them listed in Table S1. It's obvious that the specific surface area (SSA) of the CMCNFs are much higher than that of BCAs prepared by freeze drying method and it varies with the heat treatment temperature. As can be seen from Table S1, the SSA of CMCNFs-1000 is Up to $1102 \text{ m}^2/\text{g}$ and it is clear that the SSA of the prepared aerogels increases with the rise of heat treatment

temperature. Owing to the loss of hydrogen and oxygen elements in coconut matrix, the CMCNFs present a larger specific surface area and smaller density than BCAs. Nevertheless, with the increase of heat treatment temperature, the network structure of CMCNFs become relatively dense, causing the gradual increase in density (Table S1).

BCAs fabricated by freezing drying of coconut matrix and CMCNFs prepared by delicately controlled heat treatment temperatures were examined by scanning electron microscope (SEM). It shows that all obtained monolithic CMCNFs after the pyrolysis treatment can maintain the original 3D porous interconnected network structures (shown in Figs. 3 and S2). similar to the bacterial cellulose aerogels shown in Fig. 2b and c. Moreover, the CMCNFs turn to more nanofiberized with less aggregations (area circled with green oval ring in Fig. 4a, c, e and g) with increased heat treatment temperatures. Among all samples, CMCNFs-1000 shows the most uniform and rougher porous structure. However, when the heat treatment temperature is further increased to 1100°C , the carbonized fiber will be densified, the volume shrinkage is serious (Fig. S3), and the SSA decreased to $606 \text{ m}^2/\text{g}$ and the N_2 adsorption-desorption curve and pore size distribution curve of the sample are shown in Fig. S4.

Transmission electron microscopy (TEM) was used for further characterization of three-dimensional microstructure of CMCNFs (Figs. 4 and S5). It indicated that the diameter of carbon nanofibers was about

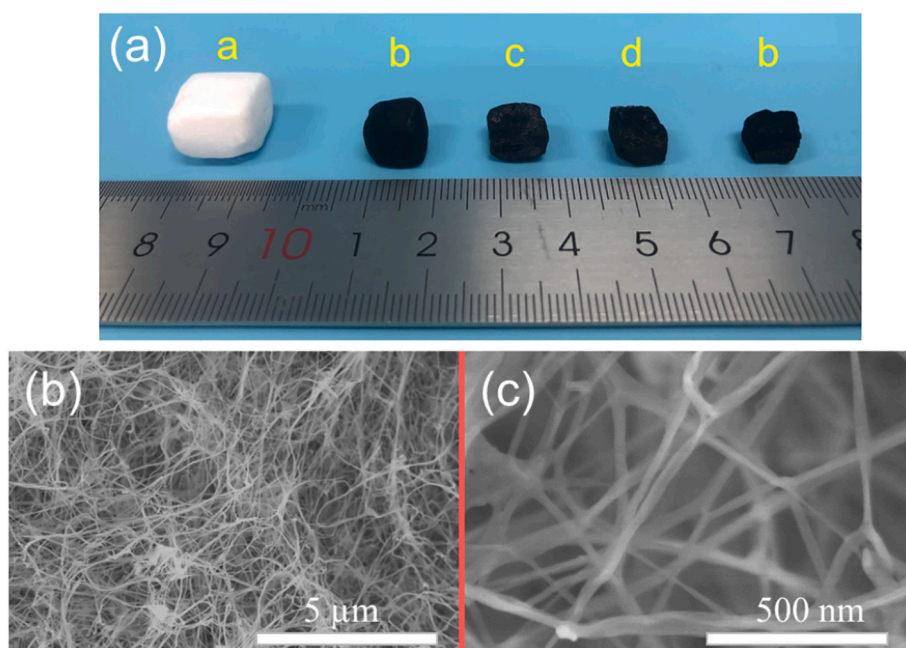


Fig. 2. (a) a: BCAs, b: CMCNFs-700, c: CMCNFs-800, d: CMCNFs-900, e: CMCNFs-1000, and (b, c) SEM images of BCAs.

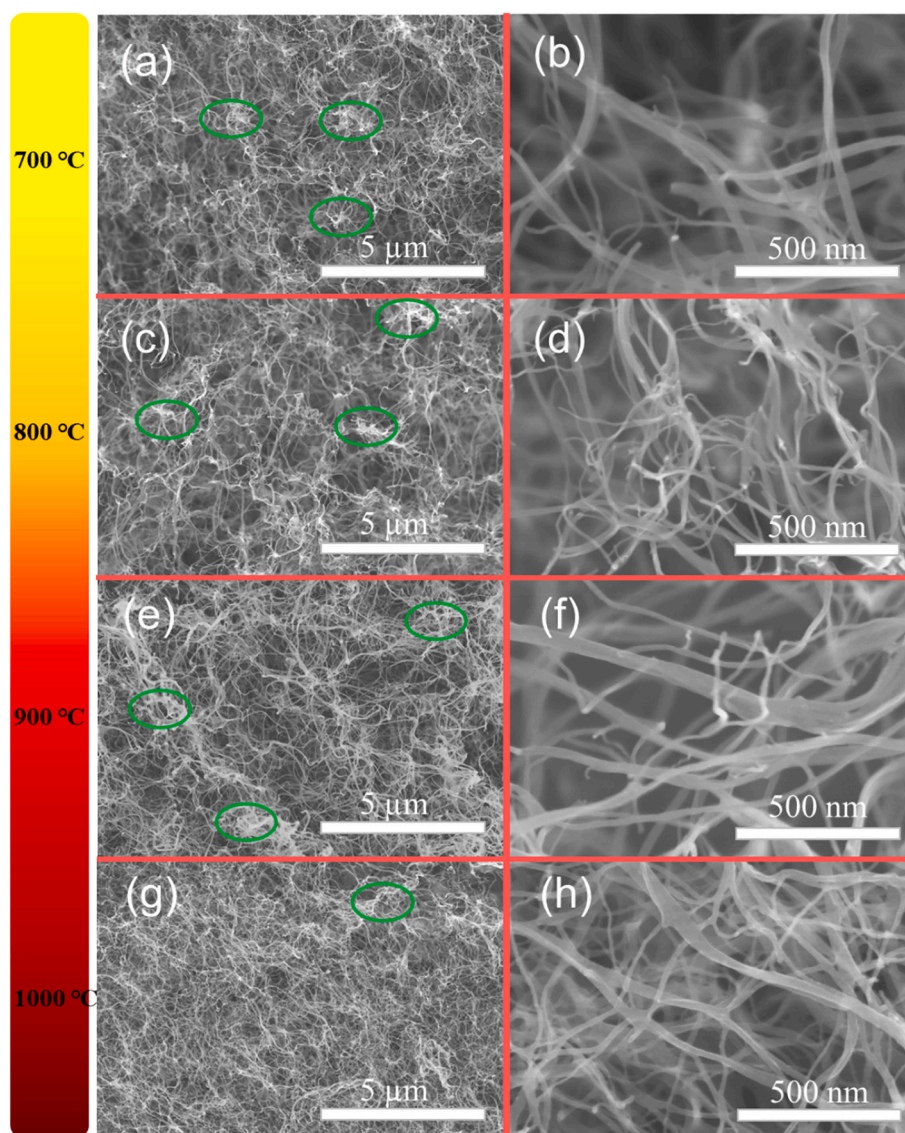


Fig. 3. SEM images of CMCNFs prepared at (a, b) 700 °C; (c, d) 800 °C; (e, f) 900 °C and (g, h) 1000 °C.

10–20 nm interconnected with many junctions to form networked mesopores. Similarly, CMCNFs-1000 (Fig. 4d) has the most uniform with more links than other samples (Fig. 4a, b, c).

3.3. SSA and pore structure

In this study, Barrett-Joyner-Halenda (BJH) method was applied to analyze the adsorption curve to study the pore size distribution more intuitively. It is shown that the pore size and pore volume of CMCNFs are affected by the carbonization temperature. As shown in Fig. 5b and Table S1, with the increase of heat treatment temperature, the pore volume increases gradually. In more details, pore volumes of 1.25 cm³/g, 3.03 cm³/g, 3.07 cm³/g, 3.65 cm³/g and 3.76 cm³/g correspond to samples of BCAs, CMCNFs-700, CMCNFs-800, CMCNFs-900, CMCNFs-1000, respectively. And the average pore size remains unchanged at about 3 nm followed by an almost flat tail covering mesopores region over 2–15 nm in diameters. Specifically, the most probable distribution values of the sample decrease gradually with the increase of the treatment temperature. This may be due to the shrinkage of the sample during high temperature heat treatment, which is a common phenomenon in the aerogel materials [49–51]. The N₂ adsorption/ desorption

isotherm from the sample can be clearly observed in Fig. 5a and the samples showed IV type isotherms and had capillary condensation apparently. It can be seen that the sample CMCNFs-1000 shows the hysteresis loop (typical characteristics of mesoporous materials), and its nitrogen adsorption capacity reaches as high as 4500 cm³/g STP, which is consistent with the large specific surface area and pore volume. Moreover, it can be inferred that SSA will be more conducive to electron transport, which will improve the sensitivity of subsequent glucose detection.

3.4. Chemical bond property

AR-FTIR spectra of the BCAs and CMCNFs prepared by pyrolysis at different temperatures are presented in Fig. 6. It is obvious that the former shows complex absorption bands of the organic polysaccharide structure. In more details, it can be clearly observed that the O–H vibration peak appeared at 3350 cm⁻¹ and 660 cm⁻¹ agree well with the vibrational peaks of O–H (650–750 cm⁻¹), which indicates that cellulose aerogels show hydrophilic characteristics. 2898 cm⁻¹ and 1605 cm⁻¹ are the vibrational peaks of C–H and 1056 cm⁻¹ can be assigned to the typical carboxylic anhydride ((RCO)₂O) with absorption peak

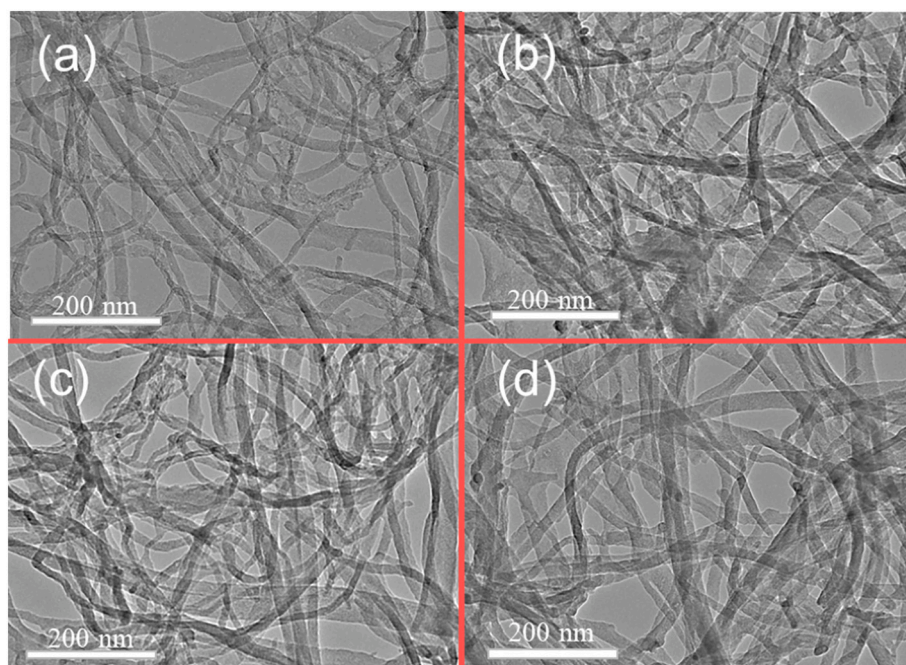


Fig. 4. TEM images of CMCNFs prepared at (a) 700 °C; (b) 800 °C; (c) 900 °C and (d) 1000 °C.

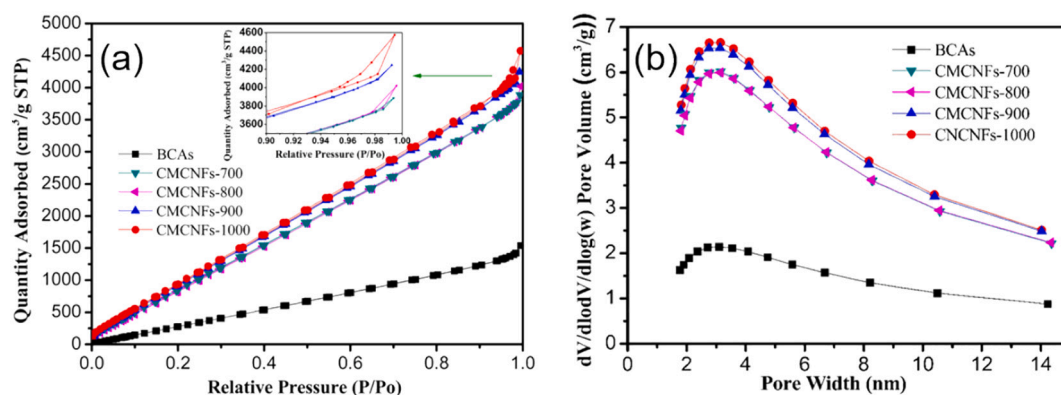


Fig. 5. (a) The N_2 adsorption-desorption curve of BCAs and CMCNFs; (b) the BJH pore size distribution curve calculated from the adsorption curve.

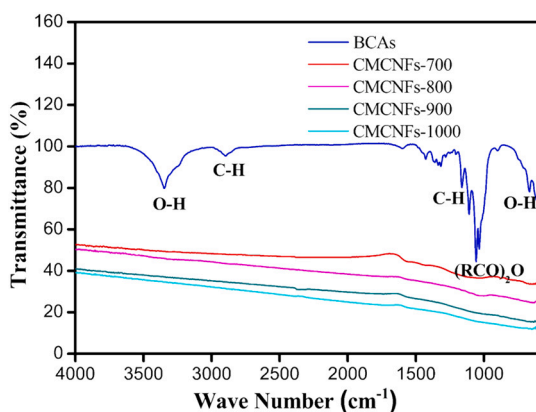


Fig. 6. Infrared diffraction spectrum of BCAs and CMCNFs pyrolysis at different temperatures.

range of (1050–1170 cm^{-1}) adsorption bands. Nevertheless, AR-FTIR absorption bands of CMCNFs show less absorption peak and weak intensity, which indicates that a large number of groups were lost during heat treatment. It should be noted that the $\text{C}=\text{C}$ stretching band of the sample obtained by carbonization treatment has a slightly blueshift. Specifically, the bands at 1605 cm^{-1} of BCAs are agree well with $\text{C}=\text{C}$ (1580–1620 cm^{-1}) adsorption bands, while the vibration peak appeared at 1582 cm^{-1} after pyrolysis treatment. In addition, Fig. S6 displays X-ray powder diffraction (XRD) pattern for CMCNFs, which consist of a broad peak and a weak peak representing the graphitic stacking of (002) and the reflections of the overlapped (101) face. And EDX of CMCNFs-1000 comprises carbon element mainly and a small amount of oxygen element (Fig. S7). These indicate that BCAs were decomposed into H_2O , CO_2 , etc., under high temperature treatment.

3.5. Direct electron transfer of GOx on CMCNFs

The electrochemical properties of CMCNFs were studied using cyclic voltammogram (CV) under conditions of 0.05 M, pH = 7.0, N_2 -saturated

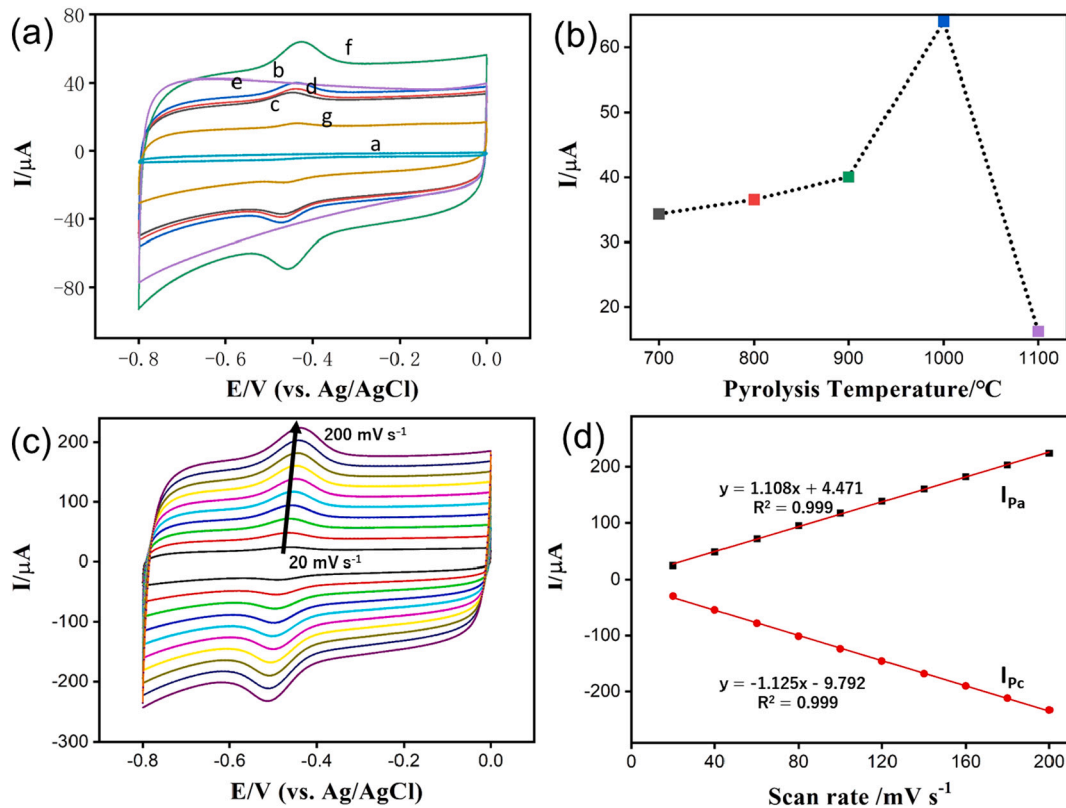


Fig. 7. (a) a: CVs obtained at GOx; b: CMCNFs-1000; c: CMCNFs-700/GOx; d: CMCNFs-800/GOx; e: CMCNFs-900/GOx; f: CMCNFs-1000/GOx; g: CMCNFs-1100/GOx modified GCEs in 0.05 M N₂-saturated PBS at the scan rate of 50 mV s⁻¹; (b) the relationship between oxidation peak current and pyrolysis temperature; (c) CVs of CMCNFs-1000/GOx modified GCEs in PBS at different scan rates from inside to outside are 20–200 mV s⁻¹; (d) plots of peak currents versus scan rate.

PBS at the scan rate of 50 mV s⁻¹. Fig. 7a displays the CV of the different modified electrodes and the direct electrochemistry between the GOx and CMCNFs can be observed from the CV response. In more details, none redox peaks are observed at GOx modified GCE in a curve indicating that direct electrochemistry of GOx is not achieved on bare electrode. Similarly, none redox peak appeared at the CMCNFs-1000 modified GCE in b curve (Fig. 7a), suggesting that CMCNFs are not electroactive in the potential range. However, the CVs of CMCNFs/GOx modified GCEs demonstrate well-defined redox peaks, of which the CMCNFs-1000/GOx delivers the largest current (Fig. 7b), indicating that CMCNFs-1000 traps the highest GOx. The plausible reasons are that samples obtained at high temperatures are fully carbonized with unique conductivity, and the high specific surface area provides more enzyme loading sites, bring about excellent electrochemical performance. In particular, the redox peaks appeared at the CMCNFs-1000/GOx

modified GCE with a formal potential ($E^{\circ'}$) of -0.442 V at pH = 7.0 (Fig. 7a), which revealed the reported value for redox potential of FAD/FADH₂ in GOx [52]. In addition, the peak potential separation (ΔE_p) was about 32 mV. The larger currents and smaller ΔE_p revealed the fast direct electron transfer of GOx (FAD/FADH₂) at the CMCNFs-1000/GOx modified GCE, due to the large surface area and favorable orientation of GOx.

The effect of scan rate on the CV response at CMCNFs-1000/GOx modified GCE was examined in N₂-saturated PBS (pH = 7) at various scan rates from 20 to 200 mV s⁻¹ as shown in Fig. 7c. Moreover, the anodic and cathodic peak currents (I_{pa}, I_{pc}) linearly increase with the increasing scan rates from 20 to 200 mV s⁻¹ (Fig. 7d), which indicates that the electron transfer of GOx is a surface control process. In addition, the electron transfer rate constant (K_s) between GOx and the electrode was calculated to 2.03 s⁻¹ using Laviron's equation as follows [53]:

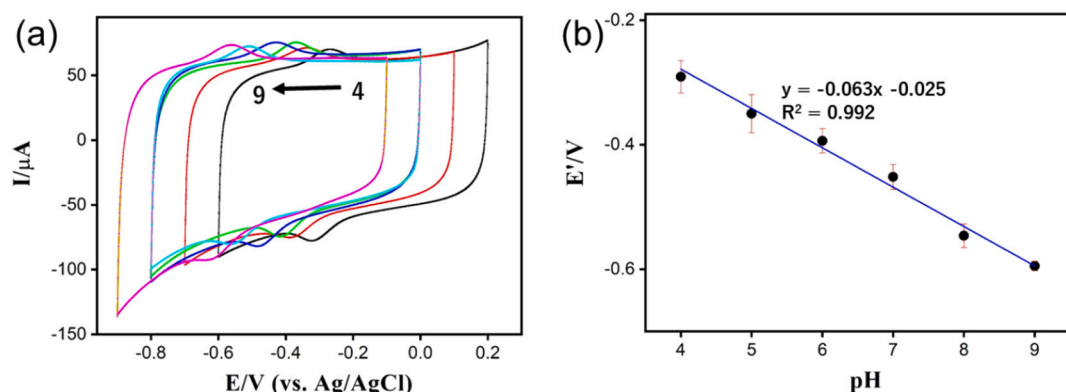


Fig. 8. (a) CVs of the CMCNFs-1000/GOx modified GCE in 0.05 M PBS with different pH values (4–9) at 50 mV s⁻¹ scan rate; (b) plot of $E^{\circ'}$ vs. pH.

$$\log k_s = a \log(1 - \alpha) + (1 - \alpha) \log \alpha - \log \frac{RT}{nFv} - \frac{\alpha(1 - \alpha)nF\Delta E_p}{2.3RT} \quad (1)$$

where α is the charge transfer coefficient (In this research, $n\Delta E_p = 64$ mV, much smaller than 200 mV, so we take α as 0.5. It was analyzed in detail in Laviron's article [53]) and the number of electrons (n) transferred (FAD/FADH₂ redox reaction) was considered as 2, ΔE_p is peak separation, the constants F , R , T have their usual meanings ($R = 8.314$ J K⁻¹ mol⁻¹, $T = 298$ K, $F = 96,485$ C mol⁻¹). In this work, the k_s value of this CMCNFs is larger than that of BPCNF/GOx (1.17 s⁻¹) [48], SWNT/GOx (0.3 s⁻¹) modified electrodes [54]. The larger rate constant suggests faster electron transfer rate between redox active sites of enzyme and electrode surface. Besides, we calculated the amount of electroactive GOx on the CMCNFs/GOx modified electrode surface (Γ) from the slope of peak currents plotted versus scan rate by the following Eq. (2) [55].

$$I_p = n^2 F^2 \nu A \Gamma / 4RT \quad (2)$$

where ν (V s⁻¹) is the scan rate and A (cm²) is the electrode surface area, the amount of electroactive GOx is calculated to be 1.25×10^{-9} mol cm⁻². The Γ value is significantly higher than that of SWNT/GOx/Nf modified GCE [54]. This result proves that the large surface area of CMCNFs facilitate the high enzyme loading.

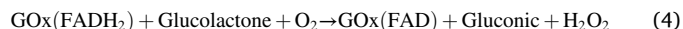
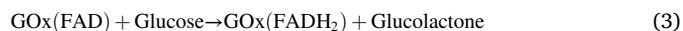
3.6. Effect of pH on modified electrode

It is well known that pH is an important parameter for the GOx based electrochemical biosensors. We studied the effect of the pH at modified electrode in various pH 4.0–9.0. GOx exhibits well defined redox peaks at different pH as shown in Fig. 8a. Moreover, the plot of pH versus $E^{o'}$ (Fig. 8b) exhibits a linear plot with pH 4.0–9.0. The linear regression equation has been expressed as $E^{o'}/V = -0.025/V - 0.063$ pH/(V/pH), while, cathodic and anodic peak potentials were shifted towards a negative shift when pH value increased. Besides the slope value of -63 mV/pH is close to the theoretical value of -58.6 mV/pH [56]. Therefore, we conclude that the redox reaction of GOx at modified electrode is reversible with the involvement of two protons (2H⁺) and two electrons (2e⁻) process [57].

3.7. Biosensing applications of CMCNFs-1000/GOx modified electrode

In order to study the electrocatalytic property of GOx adsorbed on the CMCNFs-1000, the experiments were carried out by CV in the potential range 0 to -0.8 V at the scan rate of 50 mV. Fig. S8 shows the CVs of CMCNFs-1000/GOx modified GCE in O₂-saturated PBS with different glucose concentration ranging from 0 to 6 mM (a to e). As can be seen from Fig. S8, in the absence of glucose, compared with the CV in N₂-

saturated PBS, the CV in O₂-saturated PBS has a larger reduction peak current and a smaller oxidation peak current, gradually decreased with the increase of glucose concentration. It is worth noting that the cathodic peak current of the CMCNFs-1000/GOx modified GCE in O₂-saturated condition (Fig. S8: curve a) is larger than that in the N₂-saturated PBS (Fig. 7a: curve f), which is obviously produced by the oxygen reduction of a natural co-substrate for the GOx of O₂. It confirms that oxygen dissolved in the PBS solution undergoes the reaction shown in the following equations (Eq. (3)), in which oxygen regenerates the GOx (FAD) and enhances the reduction peak current of the FAD. This process illustrates the occurrence of electrocatalysis of glucose by the GOx can detect the concentration of glucose. The possible electrocatalytic mechanism is explained by Eqs. (3) and (4).



The differential pulse voltammetry (DPV) technique was used to obtain the calibration plot and Fig. 9a displays the DPV responses of CMCNFs-1000/GOx modified GCE at different concentrations of glucose in 0.05 M O₂-saturated PBS. And the curves exhibit sharp reduction peak at -0.42 V, while the cathodic peak is decreased with increasing glucose concentrations. Moreover, it decreases linearly with increase in glucose concentrations over the range of 0.1–2, 0.2–6 mM and their respective peak currents present a linear response with the linear regression equations of $I_p(\mu\text{A}) = 95.9 [\text{glucose}]/(\mu\text{A mM}^{-1}) - 958.3/\mu\text{A}$, $R^2 = 0.985$ $I_p(\mu\text{A}) = 34.9 [\text{glucose}]/(\mu\text{A mM}^{-1}) - 883.1/\mu\text{A}$, $R^2 = 0.944$. The sensitivity and the limit of detection (LOD) of the biosensor have been

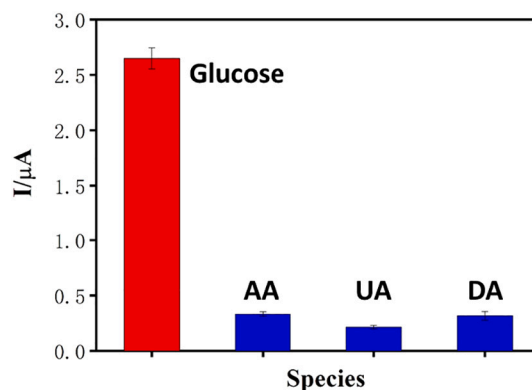


Fig. 10. Amperometric responses of AA (0.1 mmol L⁻¹), DA (0.1 mmol L⁻¹), and UA (0.1 mmol L⁻¹) and glucose (2 mM) on CMCNFs-1000/GOx modified GCE in 0.05 M O₂-saturated PBS at a constant applied potential of -0.4 V.

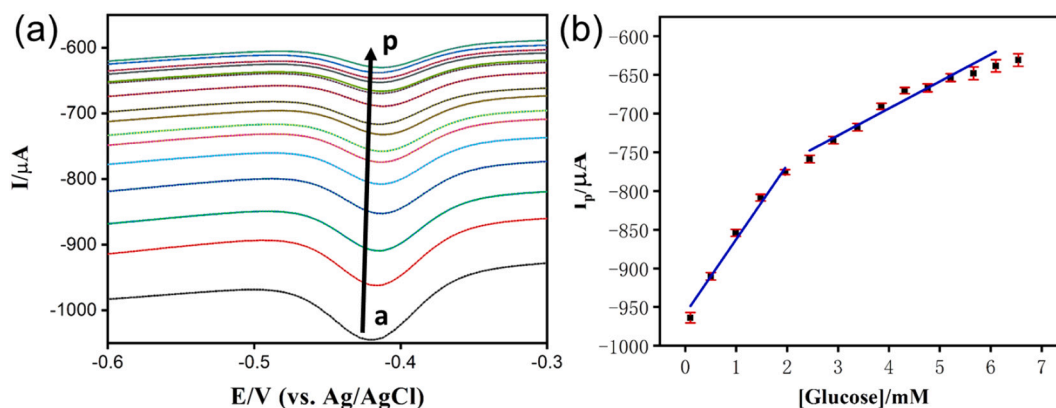


Fig. 9. (a) DPV of CMCNFs-1000/GOx modified GCE in 0.05 M O₂-saturated PBS at the scan rate of 50 mV s⁻¹ without glucose of a and with the additions of glucose 0.1–7 mM of b ~ p; (b) calibration curve of [glucose] vs. peak current. Average of three measurements (mean $\pm$ S.D.).

calculated to be $2.34 \mu\text{A mm}^{-1} \text{cm}^{-2}$ and $17.2 \mu\text{M}$ ($S/N = 3$), respectively [Fig. 9b], which is better than the GOx on GR-CNT-ZnO modified GCE previously reported [58], representing a good affinity towards glucose. We also studied the effect of glucose concentration on the amperometric response to characterize the performance of the biosensor as shown in Fig. S9. The excellent analytical parameters such as low detection limit and high sensitivity revealed the good performance of the sensor towards determination of glucose. In all, outstanding biocompatibility of the CMCNFs/GOx modified GCE shows a well electrocatalytic ability for the oxidation of glucose. The result explains that the CMCNFs-1000/GOx modified GCE is suitable for sensitive measurement of glucose.

3.8. Selectivity, repeatability and stability of CMCNFs-1000/GOx modified electrode

Selectivity is an important consideration for biosensors. The selectivity of CMCNFs-1000/GOx modified GCE was evaluated in the presence of common interfering substances (such as ascorbic acid (AA), dopamine (DA), and uric acid (UA)) in PBS. Glucose and interfering substances were consecutively added to O_2 -saturated PBS at a constant applied potential of -0.4 V (Fig. S10). In Fig. 10, no significant amperometry response to the addition of interfering substances was observed. When 2 mM glucose was added, obvious current was observed. The results showed that AA, UA, and DA did not interfere with the determination of glucose. The reproducibility of the modified electrode was investigated by CV. The concentration of 3 mM glucose was measured consecutively ten times, to investigate the reproducibility of the modified electrode. The relative standard deviation (RSD) was 2.1% , indicating a good reproducibility. In addition, the stability of CMCNFs-1000/GOx was examined through the response to 3 mM glucose as shown in Fig. S11. The results revealed that the cathodic peak current was still retained above 91% during seven days storage, indicating that a reasonably good stability.

4. Conclusion

Briefly in this study, the coconut matrix carbon nanofiber aerogels CMCNFs with highly mesoporous structure were successfully prepared through a novel method of freeze-dry and carbonized calcination ($700^\circ\text{C} \sim 1100^\circ\text{C}$) of bacterial cellulose aerogels (BCAs). The CMCNFs possess ultralight low density (density $\leq 3.47 \text{ mg/cm}^3$), high specific surface area ($\geq 1102 \text{ m}^2/\text{g}$), and high porosity ($\approx 99.6\%$) and achieve a direct electron transfer (DET) of GOx to the rapid detection of glucose. This character should be related to the three-dimensional pore structure determined by carbonization temperature, which has been proved that the CMCNFs calcined at 1000°C possesses the best capabilities. It was discovered that the good biocompatibility of the CMCNFs/GOx modified GCE exhibits a well electrocatalytic ability for the oxidation of glucose. Thus, a simple way to synthesize carbon nanofibers aerogels by using biomass is explored. The CMCNFs also has a great potential for use in enzyme glucose biofuel cells. These discovered results on the application of highly sensitive glucose detection make a significant impact for the designing of future protein-based electrochemical devices.

CRedit authorship contribution statement

Xian Yue: Conceptualization, Methodology, Validation, Investigation, Writing - original draft, Visualization. **Jiyong Feng:** Conceptualization, Methodology, Validation, Investigation, Writing original draft, Visualization. **Huaxin Li:** Conceptualization, Methodology, Software, Validation, Writing original draft, Visualization, Supervision. **Zhou Xiao:** Investigation, Validation, Visualization. **Yunsheng Qiu:** Investigation, Validation, Visualization. **Xianbo Yu:** Investigation, Validation, Visualization. **Junhui Xiang:** Conceptualization, Methodology, Validation, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.diamond.2020.108180>.

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